A Longitudinal Study of the Little Endian That Could

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What is 9002?

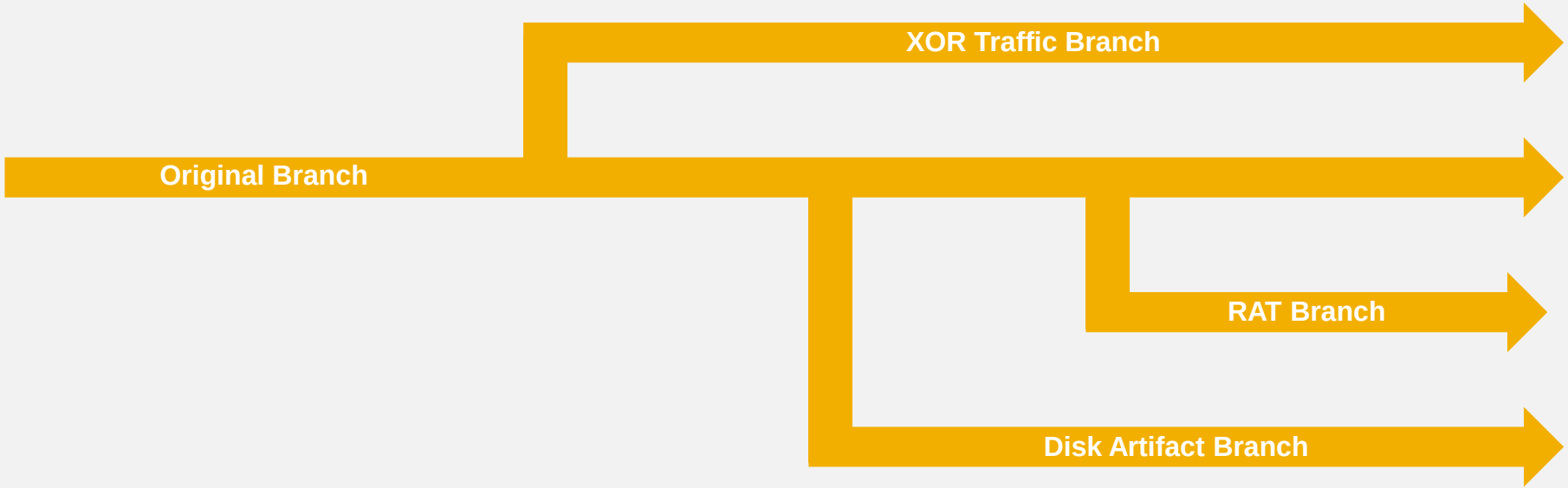
- Remote Access Tool
- Also known as:
  - NAID, Lynx, McRat, McProxy
- First seen in Operation Aurora during 2009
- Used exclusively by targeted actors
- Delivered through strategic web compromises and phishing

9002 Characteristics

- Packed code, plaintext strings
- Unusual import format
- Plugin based architecture
- “rat_Uninstall” event
- LZO compressed traffic

Expected Malware Development

9002 Malware Development



9002 Branches - Facts

- 4 unique development branches
- No overlap in improvements
- XOR Traffic Branch
  - Altered traffic format
- Disk Artifact Branch
  - Altered binaries at rest
- RAT Branch
  - Integrated plugins into main binary

XOR Traffic Branch

- Focused on modifying traffic format to evade detection
- Three major revisions
  - Two still use “Lynx” as User-Agent
- Debug strings
  - “Dog create a loop thread”
- Event uses format string
  - __rat_UnInstall__%d

Disk Artifact Branch

- Focused on modifying code at rest to evade detection
- Uses different import format
- No events registered
- No identifying strings in binaries on disk
- Configuration and malware payload stored encoded in registry

RAT Branch

- Includes keylogger plugin by default
- Updated configuration format
- Added debug strings
  - “wo try to get better”
- Uses randomly-named mutex instead of event

9002 Branches - Theories

1. Low confidence - Single source producing different versions as requested
2. Low confidence - Impersonation of existing groups as a false flag
3. High confidence - Source code shared amongst affiliated groups

Defending against 9002

- Standard threat actor playbook applies
  - Every target has a unique sample
  - Every target has a unique C2
- Hash and C2 indicators not applicable for initial detection
- Instrument everything – network, endpoint, email, etc
- Take advantage of all malware idiosyncrasies to develop countermeasures
  - Don’t rely on just one!

9002 Countermeasures

- File
  - File contains rat_UnInstall
- Network
  - Packet ends with “`19 FF FF FF FF 00 00 00 00 11 00 00`”
  - User-Agent is Lynx (assumes http proxy required)
- Endpoint
  - Event contains rat_UnInstall
  - Allocation contains %%TEMP%%\%s_p.ax

9002 Countermeasure Review

Variant	File String	Packet Sig	User-Agent	Event Name	Memory
Original-1	✓	✓	✓	✓	✓
Original-2	✓	✓	✓	✓	✓
XOR-1	✓	✓	✓	✓	✓
XOR-2	✓	✗	✓	✓	✓
XOR-3	✗	✗	✓	✓	✓
XOR-4	✗	✗	✗	✓	✓
Disk-1	✗	✓	✓	✗	✓
Disk-2	✗	✓	✓	✗	✓
RAT-1	✓	✓	✓	✗	✓

Summary

- 9002 has been in use for 6+ years
- Different development branches with different goals
  - Traffic / Disk / Features
- Hashes and C2s will not protect you against targeted threats
- Detecting intrusions requires a holistic, layered approach

Questions?

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